

NUCLEAR FUSION AND ITER

NUCLEAR-PROCESS FIREWALL → D-T ENERGY MAP → FUSION-CONDITION TRIANGLE → CONFINEMENT COMPARISON → BREAKEVEN LADDER → NEUTRON-TO-MATERIAL CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 NUCLEAR-PROCESS FIREWALL

NUCLEAR-PROCESS FIREWALL SPLITS HEAVY NUCLEI AND POWERS CURRENT REACTORS COMBINES LIGHT NUCLEI IN EXPERIMENTAL SYSTEMS COMMERCIAL STATUS FISSION OPERATING, FUSION NOT DEPLOYED SHARED WORD NUCLEAR DOES NOT ERASE PROCESS DIFFERENCES

TOPIC 05 CANNOT SUBSTITUTE FOR TOPIC 04

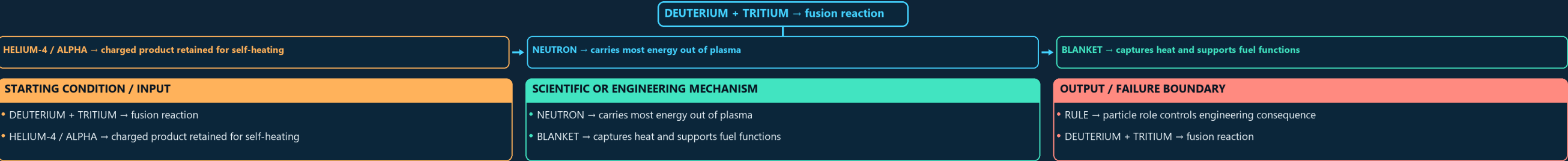
| SYSTEM / DEVICE / INSTITUTION                                                                                                                                                                          | DECISIVE TECHNICAL DISTINCTION                                                                                                                                                                             | STATUS / UNIT / EVIDENCE LIMIT                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FISSION → splits heavy nuclei and powers current reactors                                                                                                                                              | COMMERCIAL STATUS → fission operating, fusion not deployed                                                                                                                                                 | RULE → Topic 05 cannot substitute for Topic 04                                                                                                                                                  |
| FUSION → combines light nuclei in experimental systems                                                                                                                                                 | SHARED WORD NUCLEAR → does not erase process differences                                                                                                                                                   | FISSION → splits heavy nuclei and powers current reactors                                                                                                                                       |
| SYSTEM / DEVICE / INSTITUTION <ul style="list-style-type: none"><li>FISSION → splits heavy nuclei and powers current reactors</li><li>FUSION → combines light nuclei in experimental systems</li></ul> | DECISIVE TECHNICAL DISTINCTION <ul style="list-style-type: none"><li>COMMERCIAL STATUS → fission operating, fusion not deployed</li><li>SHARED WORD NUCLEAR → does not erase process differences</li></ul> | STATUS / UNIT / EVIDENCE LIMIT <ul style="list-style-type: none"><li>RULE → Topic 05 cannot substitute for Topic 04</li><li>FISSION → splits heavy nuclei and powers current reactors</li></ul> |

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → Topic 05 cannot substitute for Topic 04.

01

STAGE 01 D-T ENERGY MAP

D-T ENERGY MAP DEUTERIUM + TRITIUM FUSION REACTION CHARGED PRODUCT RETAINED FOR SELF-HEATING CARRIES MOST ENERGY OUT OF PLASMA CAPTURES HEAT AND SUPPORTS FUEL FUNCTIONS PARTICLE ROLE CONTROLS ENGINEERING CONSEQUENCE

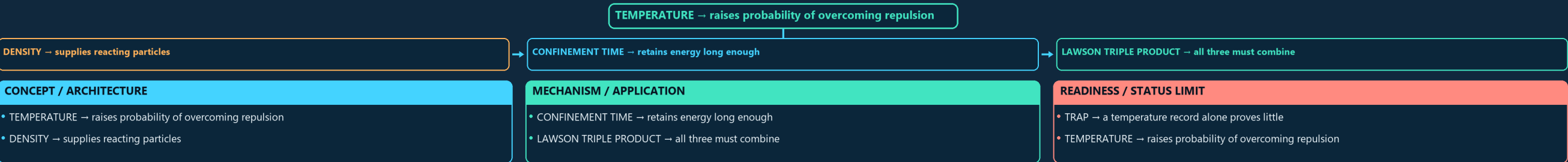


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLANKET → captures heat and supports fuel functions.

02

STAGE 02 FUSION-CONDITION TRIANGLE

FUSION-CONDITION TRIANGLE RAISES PROBABILITY OF OVERCOMING REPULSION SUPPLIES REACTING PARTICLES CONFINEMENT TIME RETAINS ENERGY LONG ENOUGH LAWSON TRIPLE PRODUCT ALL THREE MUST COMBINE A TEMPERATURE RECORD ALONE PROVES LITTLE



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → a temperature record alone proves little.

03

STAGE 03 CONFINEMENT COMPARISON

CONFINEMENT COMPARISON LOWER DENSITY AND LONGER CONFINEMENT TORUS WITH STRONG PLASMA CURRENT EXTERNALLY SHAPED STEADY-FIELD GEOMETRY DENSE TARGET AND EXTREMELY SHORT CONFINEMENT APPROACH AND DEVICE NAME ARE NOT POWER STATUS

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|-------------------------------|--------------------------------|--------------------------------|
|-------------------------------|--------------------------------|--------------------------------|

03

CONCEPT / ARCHITECTURE

- TEMPERATURE → raises probability of overcoming repulsion
- DENSITY → supplies reacting particles

MECHANISM / APPLICATION

- CONFINEMENT TIME → retains energy long enough
- LAWSON TRIPLE PRODUCT → all three must combine

READINESS / STATUS LIMIT

- TRAP → a temperature record alone proves little
- TEMPERATURE → raises probability of overcoming repulsion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → a temperature record alone proves little.

04

STAGE 03

CONFINEMENT COMPARISON

CONFINEMENT COMPARISON

LOWER DENSITY AND LONGER CONFINEMENT

TORUS WITH STRONG PLASMA CURRENT

EXTERNALLY SHAPED STEADY-FIELD GEOMETRY

DENSE TARGET AND EXTREMELY SHORT CONFINEMENT

APPROACH AND DEVICE NAME ARE NOT POWER STATUS

| SYSTEM / DEVICE / INSTITUTION                   | DECISIVE TECHNICAL DISTINCTION                          | STATUS / UNIT / EVIDENCE LIMIT                       |
|-------------------------------------------------|---------------------------------------------------------|------------------------------------------------------|
| MAGNETIC → lower density and longer confinement | STELLARATOR → externally shaped steady-field geometry   | RULE → approach and device name are not power status |
| TOKAMAK → torus with strong plasma current      | INERTIAL → dense target and extremely short confinement | MAGNETIC → lower density and longer confinement      |

SYSTEM / DEVICE / INSTITUTION

- MAGNETIC → lower density and longer confinement
- TOKAMAK → torus with strong plasma current

DECISIVE TECHNICAL DISTINCTION

- STELLARATOR → externally shaped steady-field geometry
- INERTIAL → dense target and extremely short confinement

STATUS / UNIT / EVIDENCE LIMIT

- RULE → approach and device name are not power status
- MAGNETIC → lower density and longer confinement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → approach and device name are not power status.

05

STAGE 04

BREAKEVEN LADDER

BREAKEVEN LADDER

PLASMA Q

FUSION POWER DIVIDED BY INJECTED HEATING

Q ABOVE ONE

PLASMA-LEVEL SCIENTIFIC THRESHOLD

ENGINEERING Q

INCLUDES RECIRCULATING PLANT LOADS

NET ELECTRICITY

PLASMA Q → fusion power divided by injected heating

Q ABOVE ONE → plasma-level scientific threshold

ENGINEERING Q → includes recirculating plant loads

NET ELECTRICITY → includes thermal conversion and all consumption

STARTING CONDITION / INPUT

- PLASMA Q → fusion power divided by injected heating
- Q ABOVE ONE → plasma-level scientific threshold

SCIENTIFIC OR ENGINEERING MECHANISM

- ENGINEERING Q → includes recirculating plant loads
- NET ELECTRICITY → includes thermal conversion and all consumption

OUTPUT / FAILURE BOUNDARY

- COMMERCIALITY → adds availability, cost and licensing
- PLASMA Q → fusion power divided by injected heating

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NET ELECTRICITY → includes thermal conversion and all consumption.

06

STAGE 05

NEUTRON-TO-MATERIAL CHAIN

NEUTRON-TO-MATERIAL CHAIN

14.1 MEV NEUTRON

ESCAPES MAGNETIC CONFINEMENT

RECEIVES ENERGY AND MAY BREED TRITIUM

DISPLACEMENT DAMAGE AND TRANSMUTATION

RADIOACTIVE MATERIAL-MANAGEMENT ISSUE

SEPARATE ENGINEERING FEASIBILITY CONSTRAINT

14.1 MeV NEUTRON → escapes magnetic confinement

BLANKET → receives energy and may breed tritium

STRUCTURE → displacement damage and transmutation

ACTIVATION → radioactive material-management issue

LIFETIME → separate engineering feasibility constraint

STARTING CONDITION / INPUT

- 14.1 MeV NEUTRON → escapes magnetic confinement
- BLANKET → receives energy and may breed tritium

SCIENTIFIC OR ENGINEERING MECHANISM

- STRUCTURE → displacement damage and transmutation
- ACTIVATION → radioactive material-management issue

OUTPUT / FAILURE BOUNDARY

- LIFETIME → separate engineering feasibility constraint
- 14.1 MeV NEUTRON → escapes magnetic confinement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLANKET → receives energy and may breed tritium.

STAGE 06

TRITIUM FUEL LOOP

TRITIUM FUEL LOOP

D-T PLASMA

CONSUMES TRITIUM

FUSION NEUTRON

ENTERS LITHIUM-BEARING BLANKET

LITHIUM REACTION

PRODUCES NEW TRITIUM

RETURNS USABLE FUEL

D-T PLASMA → consumes tritium

FUSION NEUTRON → enters lithium-bearing blanket

LITHIUM REACTION → produces new tritium

EXTRACTION / PROCESSING → returns usable fuel

RULE → a test blanket is not a closed commercial cycle

STARTING CONDITION / INPUT

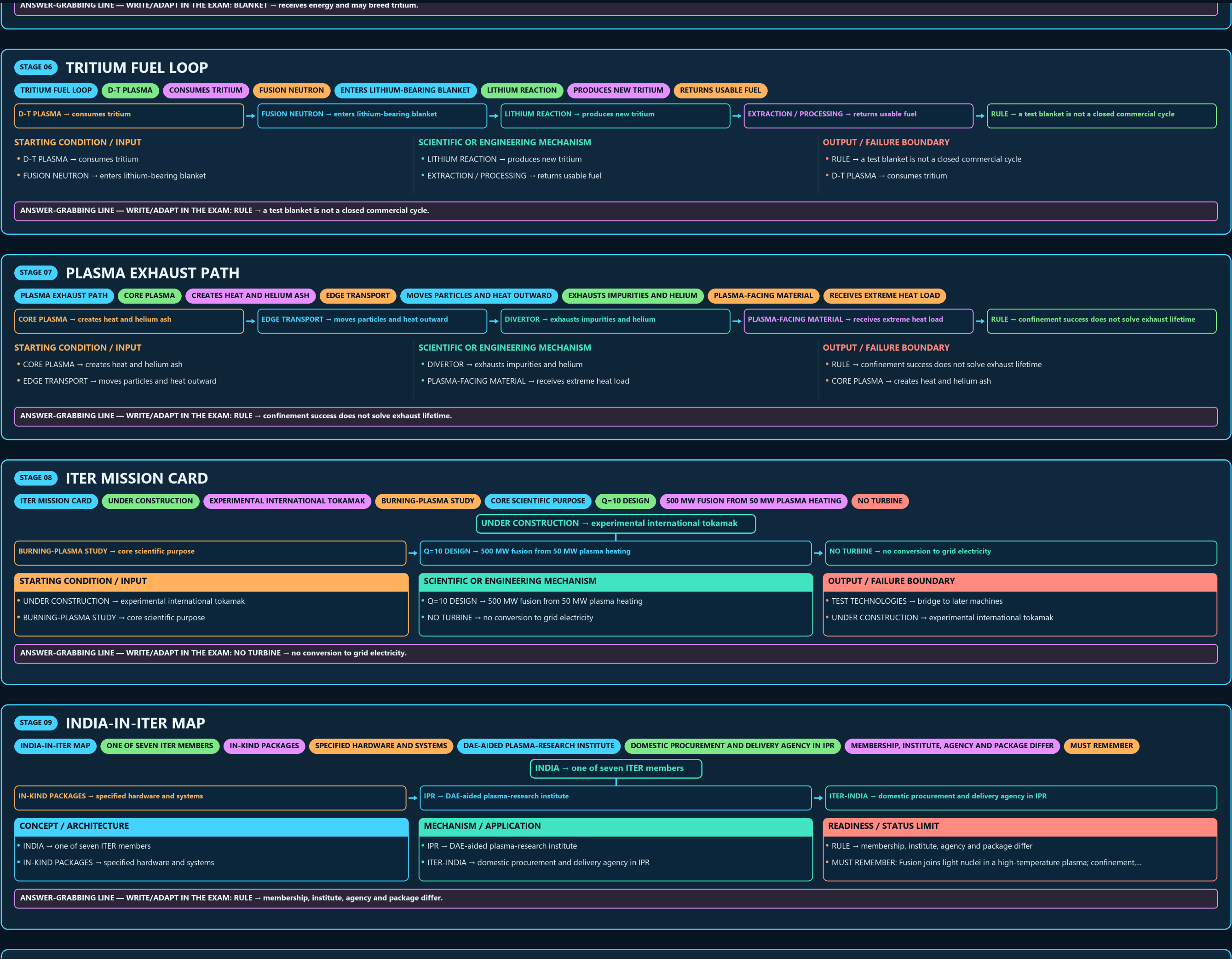
- D-T PLASMA → consumes tritium
- FUSION NEUTRON → enters lithium-bearing blanket

SCIENTIFIC OR ENGINEERING MECHANISM

- LITHIUM REACTION → produces new tritium
- EXTRACTION / PROCESSING → returns usable fuel

OUTPUT / FAILURE BOUNDARY

- RULE → a test blanket is not a closed commercial cycle
- D-T PLASMA → consumes tritium



INDIA → one of seven ITER members

IN-KIND PACKAGES → specified hardware and systems

IPR → DAE-aided plasma-research institute

ITER-INDIA → domestic procurement and delivery agency in IPR

CONCEPT / ARCHITECTURE

• INDIA → one of seven ITER members

• IN-KIND PACKAGES → specified hardware and systems

MECHANISM / APPLICATION

• IPR → DAE-aided plasma-research institute

• ITER-INDIA → domestic procurement and delivery agency in IPR

READINESS / STATUS LIMIT

• RULE → membership, institute, agency and package differ

• MUST REMEMBER: Fusion joins light nuclei in a high-temperature plasma; confinement,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → membership, institute, agency and package differ.

10

STAGE 10

DOMESTIC RESEARCH DEVICES

DOMESTIC RESEARCH DEVICES

EXPERIMENTAL TOKAMAK AND PLASMA CAMPAIGNS

SUPERCONDUCTING LONG-PULSE RESEARCH PLATFORM

RESEARCH OUTPUT

DIAGNOSTICS, CONTROL AND OPERATING KNOWLEDGE

NOT OUTPUT

COMMERCIAL FUSION ELECTRICITY

EXPERIMENTAL OPERATION IS NOT POWER GENERATION

| SYSTEM / DEVICE / INSTITUTION                        | DECISIVE TECHNICAL DISTINCTION                                 | STATUS / UNIT / EVIDENCE LIMIT                                                           |
|------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| ADITYA-U → experimental tokamak and plasma campaigns | RESEARCH OUTPUT → diagnostics, control and operating knowledge | RULE → experimental operation is not power generation                                    |
| SST-1 → superconducting long-pulse research platform | NOT OUTPUT → commercial fusion electricity                     | CLOSE DISTINCTION: Plasma Q is not engineering breakeven, tokamak is not stellarator,... |

SYSTEM / DEVICE / INSTITUTION

• ADITYA-U → experimental tokamak and plasma campaigns

• SST-1 → superconducting long-pulse research platform

DECISIVE TECHNICAL DISTINCTION

• RESEARCH OUTPUT → diagnostics, control and operating knowledge

• NOT OUTPUT → commercial fusion electricity

STATUS / UNIT / EVIDENCE LIMIT

• RULE → experimental operation is not power generation

• CLOSE DISTINCTION: Plasma Q is not engineering breakeven, tokamak is not stellarator,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → experimental operation is not power generation.

11

STAGE 11

FUSION-READINESS ANSWER SPINE

FUSION-READINESS ANSWER SPINE

D-T FUSION PLASMA CONFINEMENT LAWSON AND Q

ALPHA HEATING NEUTRON BLANKET TRITIUM AND MATERIALS

PLASMA GAIN ENGINEERING GAIN AND NET ELECTRICITY

ITER INDIA IPR ITER-INDIA ADITYA-U SST-1 AND DEMO

DATED SCHEDULE COST CONTRIBUTION AND MILESTONE EVIDENCE

EVIDENCE LIMIT

STATE QUANTITY, DEVICE, PULSE/ENERGY

DEFINE → D-T fusion plasma confinement Lawson and Q

TRACE → alpha heating neutron blanket tritium and materials

SEPARATE → plasma gain engineering gain and net electricity

MAP → ITER India IPR ITER-India ADITYA-U SST-1 and DEMO

VERIFY → dated schedule cost contribution and milestone evidence

CONCEPT / ARCHITECTURE

• DEFINE → D-T fusion plasma confinement Lawson and Q

• TRACE → alpha heating neutron blanket tritium and materials

MECHANISM / APPLICATION

• SEPARATE → plasma gain engineering gain and net electricity

• MAP → ITER India IPR ITER-India ADITYA-U SST-1 and DEMO

READINESS / STATUS LIMIT

• VERIFY → dated schedule cost contribution and milestone evidence

• EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State quantity, device, pulse/energy...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State quantity, device, pulse/energy...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State quantity, device, pulse/energy.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

ITER ORGANIZATION

GLOBAL FLAGSHIP EXPERIMENTAL FUSION COLLABORATION OF SEVEN MEMBERS —

CHINA, JAPAN, SOUTH KOREA, RUSSIA AND THE USA —

INDIA AS ITER MEMBER

CONTRIBUTES NINE IN-KIND PACKAGES — CRYOSTAT, IN-WALL SHIELDING, COOLING-WATER

SYSTEM, ION-CYCLOTRON RF HEATING, ELECTRON-CYCLOTRON RF HEATING, DIAGNOSTIC NEUTRAL

INSTITUTE FOR PLASMA RESEARCH (IPR), GANDHINAGAR

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

• ITER Organization: global flagship experimental fusion collaboration of seven members — India, EU,

• China, Japan, South Korea, Russia and the USA — sited at Saint-Paul-lez-Durance, France.

• India as ITER member: contributes nine in-kind packages — cryostat, in-wall shielding, cooling-water system, cryogenic

• system, ion-cyclotron RF heating, electron-cyclotron RF heating, diagnostic neutral beam, power supplies and diagnostics.

READINESS, UNIT OR STATUS LIMIT

• Institute for Plasma Research (IPR), Gandhinagar: a DAE-aided institute

• India's plasma-physics and fusion-research base, operating ADITYA-U and SST-1.

• ITER-India: the Indian Domestic Agency for ITER, constituted as a dedicated project within IPR .

• It contracts Indian industry, manages delivery schedules and interfaces with the ITER Organization.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

• CAUTION: Writing "IPR / ITER-India" as one entity blurs a research institute with a procurement agency — precisely

• SST-1 and Aditya-U: domestic experimental platforms that cultivate operational knowledge and scientific manpower

• IPR describes them as research devices, not power reactors.

• CAUTION: Fusion cooperation is simultaneously scientific diplomacy, industrial capability-building and long-horizon energy strategy.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.